NIAGARA 4 · API COMPARISON

niagaraRestApi v1
How it compares

Built-in XML API · **niagaraRestApi v1** · Third-party connector · Built-in REST · Analytics

Side-by-side comparison of approaches to interact with Niagara N4 stations — from the perspective of the **v1** edition.

All v1 figures verified live on a Niagara 4.14 station.

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FEATURE MATRIX

Feature comparison by channel

Feature	niagaraRestApi v1	Built-in XML	Third-party conn.	Built-in REST	Analytics
Read values	Yes	Yes	Yes	Yes	Read-only
Write values	Yes	Yes	Yes	Yes	No
Create / delete components	Yes	No	Yes	Limited	No
Deep copy (link rebuild)	Yes	No	No	No	No
Rename (link-safe)	Yes	No	No	No	No
Link / unlink	Yes	Partial	Yes	No	No
Invoke actions	Yes	Partial	Yes	No	No
Atomic batch (single POST)	11 ops	No	Limited	No	No
Facet editing	No	No	No	No	No
BQL query → JSON	No	No	No	No	Yes
Program management	No	No	No	No	No
Built-in auth (Basic)	No	Yes	Yes	Yes	Yes
CIDR allow-list	No	No	Partial	No	No
Alarm / history query	No	Yes (best)	Partial	Partial	Yes
License cost	Free	Free	Per station	Free	Subscription
SMA required	No	No	No	Yes	No
Port	8081	—	—	—	—

v1 is the free edition. It has no authentication and no BQL / facet / program features — those are v2.

WHAT V1 GIVES YOU

v1's strengths

11 batch operations

- ADD · LINK · RENAME · CP / DUPLICATE
- MOD · UNLINK · REMOVE · READ · LIST · INVOKE
- All 11 in a single atomic POST

Real introspection

- `/api/modules` — real module list with type counts
- `/api/types` — real type enumeration per module
- No credentials needed on any endpoint

Lifecycle & integrity

- Deep copy rebuilds internal links
- Rename preserves links (handle-based ORD)
- Link / unlink with auto type conversion

Drop-in simplicity

- Single `.jar`, no SMA required
- Free, no license fee
- Best for trusted networks and quick automation

ENDPOINTS

v1 HTTP endpoints

Endpoint	Method	Purpose
`/api/status`	GET	Service status
`/api/modules`	GET	Loaded modules with type counts
`/api/types`	GET	Component types for a module
`/api/config`	GET	Browse the station tree
`/api/batch`	POST	Batch component operations (11 ops)

Example — create a folder and a point

```
curl -X POST http://<HOST>:8081/api/batch \
  -H "Content-Type: application/json" -d '[
{"OP":"add","PATH":"/","T":"baja:Folder","N":"myFolder"},
{"OP":"add","PATH":"/myFolder","T":"control:NumericWritable","N":"pt"}
]'
```

Note: v1 requires no credentials — do not expose port 8081 to the public internet.

DECISION GUIDE

When to use v1

Choose v1 when…

- You are on a trusted, internal network
- You want a free, single-.jar drop-in
- You need batch component operations only
- You want real `/api/modules` and `/api/types` listings
- No SMA is available on the station

Move to v2 when…

- You need authentication and access control
- You need BQL queries returning JSON tables
- You need program management over HTTP
- You need facet editing (units, enum ranges)
- You need to configure alarm/history extensions over REST

Use the built-in XML API or Web Box for alarms and history — v1 has no such endpoints.

CONTACT

Let's build your automation

Commercial customization, integration, or deployment support.

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All niagaraRestApi v1 figures verified live on a Niagara 4.14 station. Features, availability and pricing of third-party products may vary — verify directly with each vendor.

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